

ALG-CLTAV-01 wrapper

CL-TAV specification (CR-2026-013 / DD-033)

This wrapper exists only to compile `ALG-CLTAV-01.tex`. It is not a second algorithm. The procedure is the typeset form of the method-report contract. It is not an executable engine and does not select solver technology. Step labels **S0–S10** are the stable bilingual mapping; typeset line numbers are not cited.

Chinese step groups (same process, not a second program)

- **S0**: 初始化受控会话 Γ 、历史柄 η 、资源模式、 H_0 与 retry 计数。
- **S1**: 停止 / 错误优先门；已决定的 A2–A5 或 P1–P5 立即返回。
- **S2**: 选择前预测；当前有效类 $\mathcal{C}$ 进入 **IF-SELECT-ADMIT**；预测缺口不得按剩余 $|H|$ 评分。
- **S3**: 互斥选择 **TEST minimax / Prep / Recover / Admit**；空 **TEST** 集不得 **arg min**；**Prep / Recover** 不进入 **minimax**。
- **S4**: 一次计费；重试上限不是预算耗尽。
- **S5**: 发出已准入动作并记录。
- **S6–S7**: 执行 **ERROR** 与解释 **ERROR** 汇入同一通路；不排除 H ；未知效果不能恢复成已知 q ；retry 递增，达上限先于 A5 返回 **Stop-Error**。
- **S8**: **Recover** 必须确认；未确认则 q 保持 **UNKNOWN**， η 恒等。
- **S9**: 有效观测才更新 η 与 H ；更新输出供下一轮 **S2** 预测。
- **S10**: 互斥 P1–P5；下一轮预测仍在 **S2**，不得用执行后 q 冒充预测。

Stop-645 在本增量之后表示仍未在能力/实现层关闭的具名 645 依赖义务，不是“645 文件缺失”。

Algorithm 1: CL-TAV top-level session (ALG-CLTAV-01)

Require: declared $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$; SessionContext Γ ; HistoryHandle η ; resource mode $\text{BUDGET}(c_{\text{min}}, B)$ xor $\text{ROUNDS}(K_{\text{max}})$; retry cap R_{max} ; action library; measurement uncertainty

Ensure: stop class, final H , η , session trace; singleton h_{normal} is not protocol PASS

// S0 init $H \leftarrow H_0$, $\Gamma.\text{retry} \leftarrow 0$, q UNKNOWN unless declared, η empty handles

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1 while true do
    // S1 stop/error gate before prediction
2    if IF-RESOURCE-STOP( $\Gamma, H, \eta$ ) already decided Admit A2–A5 or P1–P5 then return that stop,  $H$ ,  $\eta$ , trace;
    // S2 predict before select;  $\mathcal{C}$  feeds the select interface
3    ( $\text{pred}, q, \mathcal{C}$ )  $\leftarrow$  IF-PRED-OBS( $\Gamma, \eta, H$ , uncertainty, library);
4    if  $\text{pred}$  is PredictionGapError then return named prediction-gap; do not score empty  $\mathcal{C}$  as remaining  $|H|$ ;
5     $A \leftarrow$  currently enabled actions;
    // S3 exclusive kind; TEST minimax never applied to Prep/Recover or empty  $S$ 
6    ( $\text{kind}, t^*, S, \text{admit}$ )  $\leftarrow$  IF-SELECT-ADMIT( $A, q, \mathcal{C}, \Gamma, \eta, H$ );
7    if  $\text{kind} = \text{ADMIT}$  then
8        | return  $\text{admit} \in \{A2, A3, A4, A5\}$ ,  $H$ ,  $\eta$ , trace;
9    else if  $\text{kind} = \text{TEST}$  then
10       | if  $S = \emptyset$  then return named spec error; do not arg min empty  $S$ ;
11       |  $t^* \leftarrow$  one-step minimax on currently valid nonempty classes  $\mathcal{C}$  inside TEST-only  $S$ ;
12    else if  $\text{kind} \in \{\text{PREP}, \text{RECOVER}\}$  then
13       | keep the Prep or Recover  $t^*$  named by IF-SELECT-ADMIT; do not enter TEST minimax;
14    else
15       | return named spec error;
    // S4 charge once; retry cap  $\neq$  budget exhaustion
16    charge  $t^*$  once under the selected mode; refuse unadmitted issue;
    // S5 execute
17    ( $\text{record}, \text{execErr}$ )  $\leftarrow$  IF-EXECUTE-RECORD( $\Gamma, t^*$ );
    // S6 interpret, or skip to unified ERROR
18    if  $\text{execErr}$  is ERROR then
19       | fail  $\leftarrow$  true;
20    else
21       | ( $I_z, \text{obsErr}$ )  $\leftarrow$  IF-OBS-INTERPRET( $\Gamma, \text{record}$ );
22       | fail  $\leftarrow$  ( $\text{obsErr}$  is ERROR);
23    if fail then
24       | // S7 unified ERROR: keep  $H$ ; unknown effect is not a known  $q$ 
25       | do not exclude  $H$ ; keep  $\eta$ ; if effect unknown then  $q \leftarrow$  UNKNOWN;
26       |  $\Gamma.\text{retry} \leftarrow \Gamma.\text{retry} + 1$ ;
27       | if  $\Gamma.\text{retry} \geq R_{\text{max}}$  then return Stop-Error; this is not A5/Stop-Budget;
28       | continue after the one charge;
29    else if  $t^*$  is Recover then
30       | // S8 confirmation required; sent  $\neq$  confirmed
31       | ( $\text{confirmed}, \text{prepErr}$ )  $\leftarrow$  IF-PREP-RECOVER( $\Gamma, \eta, t^*, \text{record}$ );
32       | if  $\text{confirmed}$  is false then
33       | |  $q \leftarrow$  UNKNOWN; identity on  $\eta$  and  $H$ ; continue;
34       | else
35       | | proceed to S9;
36    if  $I_z$  is a valid observation class then
37       | // S9 history update;  $\eta'$  is the next IF-PRED-OBSinput
38       | ( $\eta, H$ )  $\leftarrow$  IF-HIST-UPDATE( $\eta, H, t^*, q, \mathcal{C}, I_z$ );
    // S10 exclusive P-stops; next prediction is S2, not a post-exec  $q$  fake
39    let  $\sigma \leftarrow$  first of Stop-Empty, Stop-Singleton, Stop-645, Stop-Equivalent, Stop-Budget via
40    IF-RESOURCE-STOP( $\Gamma, H, \eta$ ) and IF-EQUIV( $\Gamma, H, \eta$ );
41    if  $\sigma$  is set then return  $\sigma$ ,  $H$ ,  $\eta$ , and the trace;

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